Property IoT for Insurance Perils

Internal engineering roadmap · North America

Escape of water · electrical fire precursors
LoRaWAN US915 · BLE · Thread/Matter · LTE-M

14 October 2026 · **This is not an investor deck.** No market-size slide. Figures marked [V] are vendor-published; [E] are estimates.

What this deck decides

Decision	Options	Decide by
Module ecosystem	RAK/Nordic/Espressif on one SDK	**Week 1**
Radio	LoRaWAN · BLE · LTE-M · Thread	Week 6 (P0 gate)
Shutoff story	Resell certified · retrofit actuator · advisory only	Week 6 (P0 gate)
Build model	Own-design · ODM rebadge · licence · software	Month 5 (P2)

- Week-1 decision is the *ecosystem*, not the radio. Radio is downstream of the buyer.
- Everything else in year 1 is cheap to reverse. The module choice is not.

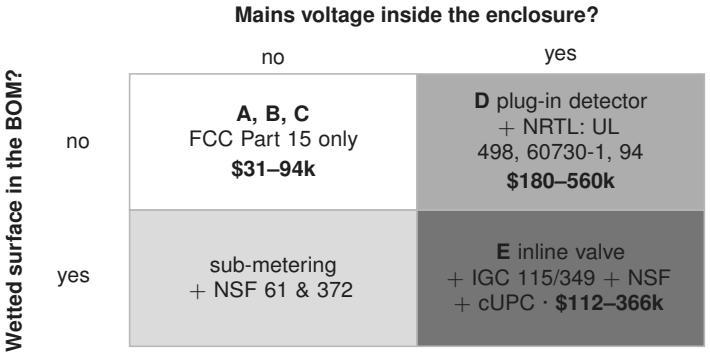
The constraint

- **$25k–$100k non-salary** for the first 12 months.
- **Full in-house EE** — schematic, PCB, RF, enclosure.

- So design labour costs **calendar**, not cash.
- What costs cash: **test labs, listings, tooling, parts, lab time** — and none of it gets cheaper because the layout is good.

Certification is a **$20–60k** problem.
The company-ending problems are **nobody will sign** and **a winter of data cannot be bought**.

The two gates that set cost



Cross *right*: +$15–40k NRTL, plus annual factory inspections.

Cross *down*: +$25–80k plumbing listing, plus a licensed plumber to install.

All eight “start now” applications avoid both gates. That quadrant is the portfolio.

Five options, re-cut on the gates

Option	Mains	Wetted	Cash [E]	Gates	Year 1?
A. Clamp-on leak sensor (battery)	–	–	**$31–94k**	FCC Part 15	**Yes**
B. Retrofit lever actuator	–	–	$70–231k	FCC; IAPMO likely N/A	**Spike**
C. Gateway on a listed adapter	–	–	low	FCC 15B/15C	**Yes**
D. Plug-in electrical detector	**Y**	–	$180–560k	+ UL 498, 60730-1, 94	No
E. Inline motorised valve	**Y**	**Y**	$112–366k	+ IGC, NSF, cUPC	No

- **B dodges the plumbing standards** — an external bracket has no wetted surface and no pressure boundary. The blocker is *mechanical*, not regulatory.
- Ship the shutoff story on a **third-party certified valve** in year 1.

What is actually on the critical path

- **Not hardware.**
- buyer signature → installed base → **one freeze/thaw cycle of labelled data** → defensible nuisance rate → security review → filing.
- FCC and PCB sit on a *parallel* path that must merely not be late.
- The one non-compressible item is **calendar time in the field across a winter**. Cash cannot buy it; parallelism cannot shorten it.

Standing rule 1: no PCB layout in the first 90 days.
Free design labour is a trap. It spends calendar — the only unbuyable resource — on a board nobody can yet validate. **Miss October and you slip a year, not a quarter.**

The spine: P0–P4, one hard gate each

Phase	Window	Cash [E]	The single exit gate
P0 Falsify the buyer	wk 0–6	$6–14k	A **countersigned** paid pilot or LOI for $\geq$50 units *with a start date*
P1 Prove the alert	wk 6–20	$12–28k	$<$**0.5 nuisance alerts/device-year** over $\geq$90 days on $\geq$50 devices, frozen eval set, $\geq$1 true event
P2 Own the node	mo 5–12	$25–55k	**FCC grant in hand**; first run at landed cost $\leq$60% of contracted price
P3 Mains & water	mo 12–24	**$150–400k raise**	Signed MSA + $\geq$1 state rate/rule filing in progress
P4 Risk data product	mo 24–36		An *approved* filing, or renewal at measurable price uplift

- P0–P2 total **$43–97k** — inside the envelope, *if* the first production run is PO-funded.
- Shaded rows **require a raise**. P3 alone is 2–4$\times$ the entire year-1 budget.

Layer × phase, layers 1–5

Layer	P0	P1	P2	P3	P4
Sensing	2×TMP117 on kit	ODM sensors	own PCB	+mains/wetted	multi-peril
Edge compute/ML	none	none (cloud)	CMSIS-DSP	emlearn on-device	on-device anomaly
Radio & WAN	LoRaWAN bench	buyer decides	one family, 2nd-src	+Matter	+Sidewalk eval
Device mgmt & OTA	none	Golioth trial	MCUboot + mcumgr/BLE	staged + canary	fleet SLA
Cloud & data	ingest + raw retention	TTS, VerneMQ, IoTDB	Magistrala + RBAC	multi-tenant portal	warehouse

- **No neural network on the node before P2.** You cannot compress a model you have not trained on data you do not have.
- **Raw retention from day one.** The corpus is the only asset cash cannot compress.

Layer × phase, layers 6–9

Layer	P0	P1	P2	P3	P4
Analytics & risk model	hand-labelled	shadow classifier	tuned rejection	severity model	portfolio score
Who signs the paper	25 convos, 4 routes	**1 countersigned**	aggregator	carrier MSA	+reinsurer
Compliance & cert	IAPMO scope letter	E&O; SOC 2 I	**FCC grant**	NRTL, IGC, NSF	SOC 2 II
Field ops & support	you install	install-verify telemetry	installer web app	dispatch SLA	regional par ners

- **"Who signs the paper" is a technology layer here**, not a sales footnote — it is the most likely single cause of death.
- Analytics is the only layer that is *actually the product*.

Build / use / partner, per layer

Layer	P0	P1	P2	P3	Why
Sensing	U	U	**B**	**B**	Commodity until the clamp mechanics are yours
Edge compute/ML	–	–	U	**B**	No NN before P2
Radio & WAN	U	U	U	U	**Never build a radio.** Modular inheritance is worth $10–25k/SKU
Device mgmt & OTA	–	U	U	U	**Never build this.** Golioth is Apache-2.0 with PKI rotation
Cloud & data	**B**	**B**	**B**	**B**	Cheap, and where carrier-facing differentiation lives
Analytics & risk	**B**	**B**	**B**	**B**	**The only layer that is the product**
Who signs the paper	**B**	P	P	P	Pre-seed cannot clear carrier vendor qualification
Compliance & cert	P	P	P	P	Labs and NRTLs, always
Field ops & support	**B**	**B**	P	P	Install quality decides your false-positive rate

B build · **U** use

existing · **P** partner · **–** not yet. Shading reinforces the token; it does not replace it.

Sensing front ends: what each can discriminate

Modality	Detects	Cannot discriminate	Parts [V]
Thermal ΔT (clamp-on)	continuous draw anywhere on the main; claimed to 5 ml/min	leak *location*; a running appliance vs a leak without time context	2$\times$ TMP117 $3.62@1k
Acoustic	pipe-borne leak noise; bursts	HVAC, traffic, appliance noise without ML	TDK T5838 (**20 µA wake-on-sound**); SPH0645 $1.45@1k
Vibration	structure-borne events	low-amplitude seepage	ST IIS3DWB $13–15@1k — **2–3$\times$ the radio**
Flow (ultrasonic)	true volume; $<$1 L/h	nothing — but it is a meter-class BOM	MSP430FR6047 $9.25@1k
Pressure decay	leaks from pressure *alone*, single sensor	needs fast sampling; a DSP claim not a sensor claim	ordinary transducer + DSP

- **Use two TMP117, not an NTC divider.** Self-heating and tolerance stack eat a small persistent ΔT.
- The T5838's **20 µA acoustic activity detect** is what makes battery acoustic viable at all.

Edge compute budget

Approach	Flash	RAM	Verdict
CMSIS-DSP features only	$<$10 kB	$<$2 kB	**P0–P2.** FFT/MFCC/filters serve water *and* electrical
emlearn random forest	**from 2 kB**	**from 50 B**	**P2.** Built-in anomaly detection; MIT
Microchip arc baseline	4–6 kB	1–2 kB	4 features, 160-sample windows @16 kHz. *Proprietary — read, reimplement clean-room*
tflite-micro + CMSIS-NN	50–250 kB	20–60 kB	P3. Only once the nuisance rate is known

On a coin cell, **a 2 kB forest that runs for five years beats a 200 kB CNN that runs for eight months. LiteRT is not the MCU successor to TF Lite** — it has no microcontroller support. Stay on `tflite-micro`.

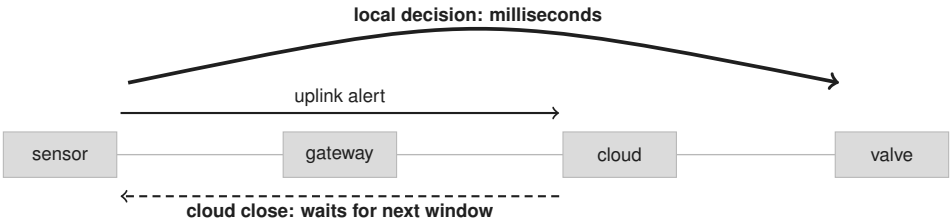
Radio decision matrix

Criterion	LoRaWAN	BLE	Thread/Matter	LTE-M	Wi-Fi	
Infrastructure you supply	gateway	phone/hub	border router	**none**	none	
Works day one, no setup	–	–	–	**Y**	–	
Needs customer Wi-Fi creds	–	–	yes	–	yes	
Battery life (sensor duty)	5–10 yr	2–5 yr	1–3 yr	2–5 yr	months	**No single right**
Basement/crawlspace reach	good	poor–mod	poor–mod	good	poor	
Recurring $/device	~0	0	0	$1–3/yr	0	
Downlink for shutoff	**bad**	good	good	moderate	good	
Consumer ecosystem	none	moderate	**Apple/Google/Amazon**	none	moderate	
Cert adder	$3k+$1.2k	–	$10.5k	PTCRB	–	

radio. Segment by buyer: multifamily → LoRaWAN + building gateway · carrier-mailed single-family → LTE-M · DTC → BLE/Matter · mains → Wi-Fi/Thread.

The downlink problem

Class	Downlink behaviour	Consequence for a shutoff command
A	Receive window opens *only* after an uplink	At a 15-min heartbeat, worst case is **15 minutes of flooding**. Unusable
B	Scheduled ping slots on gateway beacons	Seconds, but beacon tracking burns power continuously
C	Receive window open ~continuously	Latency fine, **power budget gone** — effectively mains



Architectural, not operational: never put the shutoff decision on the WAN. LoRaWAN carries alerts uplink. The close is decided locally or inside the valve; the cloud confirms and reports.

Power and battery arithmetic

- **Li-SOCl$_2$ with a hybrid-layer capacitor. Never coin cells** for the main SKU — Li-SOCl$_2$ cannot serve the TX pulse alone.
- **Derate datasheet capacity by ~40% for a cold basement**, and *measure* pulse behaviour at -10 to 0°C rather than modelling it.
- Benchmarks to beat [V]: Milesight EM300-SLD $>$**10 yr at SF7 US915**; WS303 ~**5.7 yr** on a CR2450 590 mAh.

- **Valve actuation is the power cliff.** Moen's Flo — the market leader — requires a nearby AC outlet and sells a 25-ft extension rather than going battery.
- So: **battery-only shutoff requires a latching solenoid, not a motor.** A tens-of-ms pulse holding state at zero standing current is the only topology that gets years on cells.
- **Fail-as-is, never fail-closed.** A valve that closes on battery death causes the freeze damage you were hired to prevent. Latching is inherently fail-as-is. Mechanical override on every unit.

Dev kit → production module

Family	Prototype [V]	Production [V]	Radios	US915	Cert pack
RAK WisDuo	RAK3272S $15.00	**RAK3172 $5.99**	LoRaWAN A/B/C	**Y**	**FCC, ISED, CE, LoRa Alliance — direct downloads**
RAK WisBlock	RAK19007 + RAK4631 **$28**	RAK4631 $17.99	LoRa + BLE 5.0	**Y**	CE/FCC/ISED/KC/RCM
RAK (one die)	RAK11722 $15.00	**RAK11720 $7.99**	**LoRa + BLE**	?	unverified
Seeed/ST	Wio-E5 kit $26.99	**Wio-E5 $6.49**	LoRaWAN	**Y**	CE-RED, FCC, IC
ST bare	NUCLEO-WL55JC1 $53	STM32WLE5CCU6 **$4.17@10k**	LoRa	**Y**	**none — you own the filing**
Espressif	C6-DevKitC-1 ~$9	**ESP32-C6-WROOM-1 $2.85–3.40**	Wi-Fi 6, BLE 5.3, 802.15.4	n/a	**482-doc cert index**
Nordic	nRF52840-DK $32.87	Raytac MDBT50Q $6.92–9.45	BLE 5, Thread, Matter	n/a	FCC/IC/CE per module
Nordic cellular	nRF9151-DK $130	nRF9151 SiP	LTE-M, NB-IoT, NTN	n/a	“certified for global operation”

Decision: prototype RAK19007+RAK4631 ($28) · ship RAK3172 ($5.99) or RAK11720 ($7.99) · second-source Wio-E5 ($6.49), same STM32WLE5 die, different vendor · ESP32-C6 for anything mains.

Why the module choice is irreversible

- **Modular inheritance is worth ~$10–25k and 6–10 weeks per SKU** against a full intentional-radiator campaign.
- **Bare silicon costs more than it saves.** STM32WLE5 at $4.17@10k vs RAK3172 at $5.99: the $1.80/unit saving is ~$18k at 10k units — roughly what the extra certification costs, with all the risk on your side.
- **Module grants carry antenna type and trace conditions.** Violate them and you forfeit modular approval and owe a full Part 15C test. *Read the grant before the layout.*
- **A different module is a different FCC grant.** A supply disruption therefore costs a test campaign and 4–10 weeks of lab queue, not a part change — at exactly the moment there is no cash.
- **Three radios $\times$ three products = nine filings.** Standardising on one module family is worth five figures.

Mitigation: pre-identify a footprint-compatible alternate, **budget the re-test as a known line item**, and hold 12 months of stock after the PO.

Firmware, OS and licences

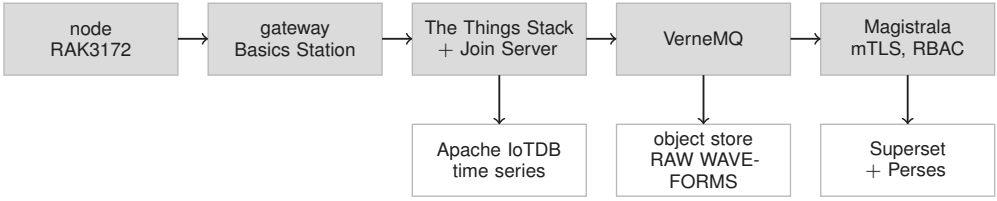
Layer	Pick	Licence	Note
RTOS	Zephyr + `west`	Apache-2.0	BLE + LoRaWAN + Matter in one tree; survives a BOM change
BLE	Zephyr BT / NimBLE	Apache-2.0	Avoid Nordic's SoftDevice blob
LoRaWAN MAC	`subsys/lorawan` over loramac-node	BSD-3	**No Apache-2.0 MAC exists.** `US915` present
Secure boot/OTA	MCUboot + mcumgr-over-BLE	Apache-2.0	Rollback; recover a brick in a basement with no WAN
Fleet mgmt	Golioth SDK	Apache-2.0	+ hawkBit (EPL) as self-host fallback
Network server	The Things Stack	Apache-2.0	**Join Server in-tree — you hold your own root keys**
Broker	VerneMQ (+ NanoMQ at the edge)	Apache-2.0	
Time series	Apache IoTDB	Apache-2.0	
App platform	Magistrala	Apache-2.0	mTLS, ABAC/RBAC, audit logs
Edge DSP/ML	CMSIS-DSP, CMSIS-NN, emlearn	Apache/MIT	CMSIS-DSP is the real workhorse
Dashboards	Apache Superset + Perses	Apache-2.0	

Blacklisted from the product path: ThingsBoard and EMQX (**BUSL-1.1** — ThingsBoard caps free commercial use at **100 devices**) · Grafana and OpenRemote (AGPL) · ESPHome (**device-side C++ is GPLv3**) · Tasmota, IotaWatt, Meshtastic, zigbee2mqtt (GPL) · LeakDB, BattLeDIM, EPyT (**EUPL-1.2 is copyleft**). All 89 verified in `data/repos.csv`.

Device management and OTA: buy it

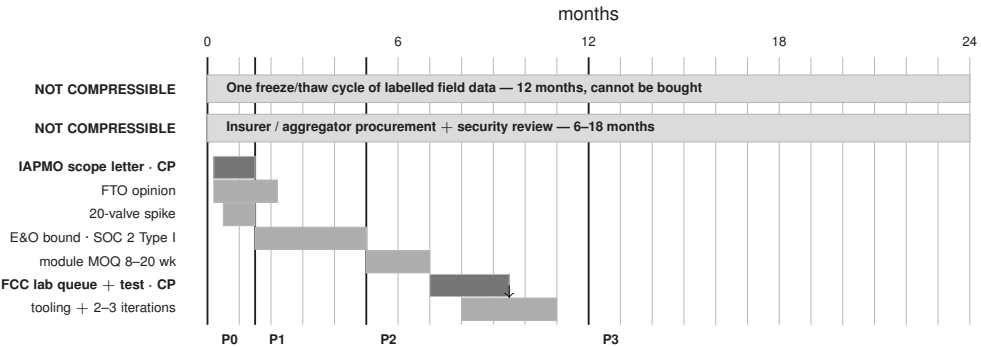
- **Never build this layer.** Golioth is Apache-2.0 with OTA, RPC and PKI certificate rotation, CI-tested on Zephyr, ESP-IDF and NCS. hawkBit (EPL) is the self-host escape hatch — fine as an unmodified standalone server you never link into firmware.
- **MCUboot for the trust format; mcumgr-over-BLE for the transport.** The BLE path is how a technician re-flashes a bricked sensor in a basement with no WAN at all — worth far more than it looks the first time a 1,000-unit batch ships with bad firmware.
- **Signed images with rollback are the difference between a bad OTA being an incident and being a mass-flooding liability event.**
- **FUOTA over LoRaWAN is a trap** until there is no alternative — unicast updates at DR0 blow the airtime budget. `chirpstack-fuota-server` (MIT) exists when you need multicast.
- Reject **SUIT**: essentially no open tooling exists. Use MCUboot's image format.
- Memfault is excellent and is **not open source** — a vendor licence shipping inside your binary. Fine as a decision; fatal on a slide labelled "our open-source stack" during diligence.

Cloud and data



- ~**$95–115/month** self-hosted at 10,000 devices. No Kubernetes, no Kafka, no ThingsBoard, no Grafana.
- **The highest-leverage spend above $200/mo is not compute — it is raw-waveform object storage.** Keep every waveform; the corpus is the input to everything.
- Spend the slack on managed Postgres for the Join Server: **losing LoRaWAN root key material is an extinction event**.

Certification critical path



CP = critical path (darkest fill). The two NOT COMPRESSIBLE bars span the whole chart; FCC and tooling are *short bars inside them*. That is the argument of the entire roadmap in one picture: **hardware is not the critical path** — it only has to not be late.

Can we dodge the plumbing standards?

Standard	Scope	Retrofit lever actuator
NSF/ANSI 61	what leaches into drinking water	**Out** — no wetted surface
NSF/ANSI 372	$\leq$0.25% lead on potable-contact surfaces	**Out** — no wetted surface
IAPMO IGC 115/349	automatic leak detection / supply-integrity devices	**Probably out** — no pressure boundary, so most of the test matrix is physically inapplicable. *Get it in writing*
UL 498 / 60730-1	mains-powered devices	**Out** if battery or an externally supplied listed adapter

- **ASSE 1087 is the wrong document** — it is commercial/food-service water *treatment* equipment. The IGC numbers above are the real ones.
- **Hard BOM rule: actuator only, no plumbing parts in the box.** One adapter and the kit becomes a plumbing product.
- The blocker is mechanical: breakaway torque on aged valves, **stem weep after years of set packing**, and **multi-turn gate valves that a quarter-turn actuator cannot drive at all**.

Exit gate. $>$**70% of twenty salvaged aged valves cycle without stem weep.** Cost: a few hundred dollars. Fail $\rightarrow$ kill option B permanently and resell a certified valve.

Risk register, top 12

	Risk	L	I	Leading indicator	Mitigation
1	**No buyer signs with a pre-seed vendor**	H	**fatal**	no countersigned doc by wk 6	route via MGA/aggregator; run all 4 routes
2	False positives destroy insurer trust	H	H	nuisance rate not falling wk/wk	shadow mode; install-verify telemetry
3	Loss reduction unprovable at this scale	H	H	pilot $<$2,000 house-years	**agree a surrogate metric in P0**
4	Install-quality failure tail	H	M	$>$10% never reporting at day 7	don't count unverified installs
5	LoRaWAN downlink unreliable for shutoff	H	H	any doc with cloud closing the valve	**architectural: decide locally**
6	Premium credit unfilable / anti-rebating	M	**fatal**	counsel won't commit to a filing path	**sell loss control, not a discount**
7	Battery life in a cold basement	H	M	pulse behaviour at -10 to 0°C	Li-$SOCl_2$ + HLC; derate 40%; measure
8	Module EOL invalidates the FCC grant	M	H	vendor PCN; lead time $>$16 wk	same-vendor alternate; budget the re-test
9	Valve fails closed (or silently open)	M	H	any unit arming above torque threshold	refuse to arm; **fail-as-is**; override
10	The data moat is not defensible	M	H	model flat per 1k device-months	own the relationship; fix data rights in contract
11	ODM dependency / landed-cost swing	M	M	landed cost moving $>$15%	own firmware + fixtures; dual-quote a NA CM
12	**Seasonality makes slip non-linear**	M	H	devices not in field by October	**a 6-week slip costs 12 months**

Likelihood and impact are text tokens, sorted by L$\times$I — deliberately not colour-coded, because red and green are ~1.3:1 apart in greyscale.

First 90 days, and what we will not build

The first 90 days

- **Wk 1:** lock the *module ecosystem* (not the radio); send the **IAPMO scope letter**; ingest live with raw retention.
- **Wk 1–2:** commission the **FTO opinion**.
- **Wk 2–6: 20–30 dev-kit nodes into 10–20 real buildings**.
- **Wk 2–12: 25 buyer conversations, all four routes** — target one countersigned document.
- **Wk 6–12:** the **20-valve spike**; bind E&O; start SOC 2 readiness.
- **Throughout: no PCB layout. No Gerbers exist.**

What we will not build

- No inline valve in year 1 — resell a certified one.
- No mains-powered product in year 1.
- **No arc-fault interruption, ever** — that is UL 1699 at +$60–200k. Alert only, like Ting.
- No own device-management or OTA backend.
- No neural network on the node before P2.
- No FUOTA over LoRaWAN; no own network server under 1,000 nodes.
- No Matter before a DTC buyer exists.
- **No AGPL or GPL in the product path.**
- No custom gateway hardware — RAK sells them from $99.
- No premium-share or MGA structure in year 1.

Exit gate. P0: a countersigned paid pilot or LOI for $\geq$50 units *with a start date*.

Appendix A — indicative BOM, clamp-on sensor

Item	Part	$ @1k	Evidence
Radio + MCU module	RAK3172 (STM32WLE5)	5.99	[V]
alternate, adds BLE	RAK11720	7.99	[V]
second source	Seeed Wio-E5	6.49	[V]
Pipe temperature	TI TMP117	3.62	[V]
Ambient temperature	TI TMP117	3.62	[V]
Tamper/sanity channel	NTC	0.15	[E]
Battery	2× Li-$SOCl_2$ AA	5.00	[E]
Pulse support	hybrid-layer cap	1.20	[E]
Antenna	chip/wire 915 MHz	0.45	[E]
PCB + assembly	4-layer 40×60 mm	3.50	[E]
Enclosure, clip, thermal pad	moulded ABS/PC	4.00	[E]
Total		**~27**	mid-point

Excludes amortised tooling, freight, duty and test. Full file: `data/bom-estimates.csv`.

Appendix B — certification line items

Line (clamp-on sensor)	Low	High	Evidence
FCC/ISED composite, module grant inherited	1,500	5,000	[E]
LoRa Alliance membership (startup rate)	3,000	6,000	[V]
LoRaWAN cert, end device at an ATL	1,200	1,700	[V]
Plumbing listing	0	0	verified
NRTL safety listing	0	0	verified
Tooling, EVT/DVT, FTO opinion	25,000	80,000	[E]
Total to shippable	**~31k**	**~94k**	

- Plug-in detector: **$180–560k** — NRTL $15–40k, classifier $150–500k, +$60–200k if it ever interrupts.
- Inline valve: **$112–366k** — IGC + NSF + cUPC + UL 60730.
- **OSHA recognises 21 NRTLs.** Get three quotes — Intertek ETL and QPS routinely undercut UL. NSF and IAPMO EGS are themselves NRTLs, so one body can scope both.

Appendix C — gateway economics

Deployment	Gateway	$ [V]
Single home, customer internet	RAK7266 Soho Lite	99
Single home, better radio	RAK7268V2 Edge Lite 2	154
Unreliable/absent internet	**RAK7268CV2-V (Verizon)**	**299**
Multi-dwelling	**RAK7289V2 Edge Pro**	**382**
Campus / outdoor	RAK7285 / RAK7393	799 / 439

	Single-family	40-unit building
Gateway	$99–299	$382
Nodes	$1 \times \$27$	$40 \times \$12 = \$480$
Gateway as % of HW	**79–92%**	**44%**

- **In a single-family home the gateway costs 3–10$\times$ the sensor.** That kills a consumer price point and is dead weight in an insurer-funded BOM.
- So if you ship a gateway per home anyway, **ask what LoRa is buying you** — BLE or LTE-M does it more cheaply. **LoRaWAN's economics live in buildings.**
- The $299 Verizon unit is the exception worth considering: it removes the homeowner's router from your support burden, and support cost often exceeds the $145 delta.

Appendix D — public LoRaWAN coverage reality

- **Senet no longer exists independently** — redirects to Netmore, which **acquired Actility in January 2026**. Two of three names in the original brief are now one vendor.
- **Everynet** fails its TLS handshake. **Helium** has pivoted cellular-first.
- **TTN:** 249,612 members, but ~43 forum posts in the last 7 days.
- **Amazon Sidewalk** is the dark horse — the only 900 MHz network whose gateways (Echo, Ring) are *already inside American homes*. Severe payload limits; total Amazon dependency.

The decisive argument is **operational, not coverage**: **no US operator gives you an API to pre-qualify a customer's address before you ship them a box.** A hardware company shipping devices that might not connect has an unmanageable returns cost.

Also dead: **Arm Mbed OS reached EOL in July 2026** (`os.mbed.com` now redirects to GitHub) · ESP-IDF v5.2 branch EOL August 2026 · TI AMC130M03 has no distributor stock anywhere.